EXP.NO:18

DP - LONGEST COMMON SUBSEQUENCE

PROBLEM STATEMENT:

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1	a	g	**g**	**t**	**a**	**b**	
s2	**g**	x	**t**	**x**	**a**	**y**	**b**

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab azb	2

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CODE

```c
#include<stdio.h>
#include<string.h>

int main(){
    char s1[1000], s2[1000];
    scanf("%s", s1);
    scanf("%s", s2);

    int m=strlen(s1);
    int n=strlen(s2);
    int dp[m+1][n+1];

    for(int i=0;i<=m;i++)
        dp[i][0]=0;
    for(int j=0;j<=n;j++)
        dp[0][j]=0;

    for(int i=1;i<=m;i++){
        for(int j=1;j<=n;j++){
            if(s1[i-1]==s2[j-1])
                dp[i][j] = dp[i-1][j-1] + 1;
            else
                dp[i][j] = (dp[i-1][j]>dp[i][j-1]) ? dp[i-1][j]:dp[i][j-1];
        }
    }
    printf("%d\n", dp[m][n]);
    return 0;
}
```

OUTPUT:

	Input	Expected	Got	
✔	aab azb	2	2	✔
✔	ABCD ABCD	4	4	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

RESULT:

The code is executed successfully.